# Assignment 2: Paper Reading and Summarization

## Objective

The primary goal of this assignment is to make you familiar with the structure of a research paper. It also enables you to read, summarize, and extract important key points from research papers.

## Tasks

1. Extract the following information from each paper and present in in a tabular form. Dataset, image resolution, augmentation (Y/N), task type (classification, segmentation, detection etc.), DL/ML model, novelty, loss function, evaluation metric, eval score
2. Select one paper (0-9) which matches the last digit of your student-id and do the following:
   a. Select one specific baseline paper from the Literature Review that the authors claim their proposed model outperforms.
   b. Every data preprocessing step removes some information from the raw data. Look at the authors' preprocessing pipeline. What is one specific, potentially useful piece of information or edge-case feature that is destroyed or masked by their preprocessing steps?
   c. Imagine you were tasked with coding this exact model architecture from scratch by yourself this weekend. Based on the Methodology/Architecture section, point to the single most complicated or computationally expensive mathematical operation or layer.
   d. Choose one graph, chart, or visual figure in the Results section. Identify a specific data point, spike, dip, or visual trend in that figure that the authors did not explicitly explain in the main text or the caption.

## Deliverables

Submit your assignment as a word/pdf document containing the summary table (Task 1) and the answer to question (Task 2: a-d) w.r.t your chosen research paper.

Any evidence of plagiarism or AI interference will result in ZERO points.